

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 13

Measurement: How the World Gets Numbers

Lecture

Self-Study Curriculum

Chapter 1, Exercise 3 asked when averaging is a meaningful operation and when it fails. You have half the answer already. Chapter 11 gave you the *statistical* half: averaging works when contributions concentrate, and fails when they are strongly dependent or heavy-tailed. This chapter gives you the other half, and it is the more fundamental of the two, because it applies *before any statistics run at all*. It is about how the world acquires numbers in the first place, and about which operations on those numbers mean anything once it has. The statistical failure is about whether your average is *stable*. The failure this chapter describes is worse: it is about whether your average is even *asking a question the world can answer*.

Notice where this sits. The whole of Phase III taught you to see mathematical objects through their structure-preserving maps. The next chapter will tell you that a model itself is such a map, from a system to a mathematical structure. But every one of those maps presupposes that the system is *already* described in numbers. Where did the numbers come from? Something had to attach them to the world before any equation could touch it. That something is measurement, and it is the first structure-preserving map in the entire pipeline, earlier than every map you met in Phase III.

The first homomorphism

Here is the whole chapter compressed into one sentence, and it is worth reading twice: *a measurement scale is itself a homomorphism*.

Unpack it with the vocabulary you now own. Representational measurement theory says that out in the world there is an **empirical relational structure**: a set of objects together with the relations among them that you can *actually observe*, without numbers, using only physical operations. This rod is longer than that rod (you lay them side by side and look). These two rods laid end to end balance that third rod (you build them and check). This solution is more acidic than that one (an indicator changes color). None of these observations mentions a number. They are relations you can establish by hand, in the world, with objects.

A scale is a map from that empirical structure into a **numerical relational structure**: the real numbers, carrying their own relations, less-than and plus and times. And the map is not free labeling. It must *preserve the observable relations*. If rod a is empirically longer than rod b, then the number assigned to a must be larger than the number assigned to b. If concatenating a and b (laying them end to end) balances c, then the number for a plus the number for b must equal the number for c. The empirical relation “longer than” must land on the numerical relation “greater than”; the empirical operation “lay end to end” must land on the numerical operation “add.” A map that carries relations to matching relations, and operations to matching operations, is exactly a **homomorphism**, the structure-preserving map at the heart of all of Phase III. Assigning numbers to the world is the construction of one.

This reframing does real work. It says the numbers in your spreadsheet are not primitive data; they are the *output* of a map whose faithfulness is a claim about the world, and that claim can be right or wrong. When length works so well that you forget there was ever a choice, it is because the empirical structure of rigid rods is so rich (you really can order them, and really can concatenate them, and the two operations really do agree the way numbers do) that the homomorphism into the reals is tight and almost forced. When measurement gets hard, in psychology, in ecology, in clinical grading, it is because the empirical structure is thin: you can observe *order* among symptoms but you cannot physically *concatenate* two mild cases to balance one severe case, so the operation “add” has nothing in the world to correspond to, and any numbers you write down that pretend it does are decoration, not measurement.

The presence or absence of a physical concatenation operation is in fact the deepest fault line

in the whole subject, and it has a name. A quantity for which you *can* physically combine objects and have the combination behave like addition is called **extensive**: length (lay rods end to end), mass (put objects on the same pan), charge, volume, count. For extensive quantities the empirical structure hands you an “add” operation for free, so the homomorphism can preserve addition, and you land at or near a ratio scale almost automatically. A quantity for which no such concatenation exists is called **intensive**: temperature (put two 20-degree bodies together and you do not get a 40-degree body), density, pH, a pain rating. Intensive quantities have to be measured *indirectly*, by riding on some extensive proxy (the length of a mercury column, the count of protons in solution) or by pure ranking, and that indirection is exactly why they so often top out at interval or ordinal rather than ratio. So the scale type of a quantity is not a labeling convention; it is a readout of how much empirical structure the world actually offers for that quantity. When you find yourself unsure whether something is ratio or merely ordinal, the operational question to ask is concrete: *is there a physical operation that combines two specimens the way addition combines two numbers?* If yes, you likely have a ratio scale and every operation is licensed; if no, you should suspect you are higher up the ladder than your spreadsheet’s tidy numbers suggest.

Scale types: how much structure the homomorphism carries

Since a scale is a homomorphism, the natural question is the Phase III question: how *rigid* is it? How many different number-assignments would preserve the *same* empirical structure equally well? A scale that pins the numbers down tightly carries a lot of structure; a scale that leaves the numbers loose carries little. The precise handle on this is the set of transformations you may apply to the numbers *without destroying any of the relations the scale encodes*. That set is the scale’s group of **admissible transformations**, and yes, it is a group in the exact sense of Chapter 7: apply one admissible transformation, then another, and you get a third; every one has an inverse that undoes it; doing nothing is the identity. The invariance thesis of Chapter 5 is about to earn its keep in the most practical possible way.

This gives Stevens’s hierarchy, four rungs, each defined by which relations it preserves and therefore by how large its group of admissible transformations is:

- **Nominal.** Numbers are pure labels: species IDs, gene names, plate-well positions. The only relation preserved is *equality* (same or different). The admissible transformations are *any* one-to-one relabeling, the largest group of all. You may recode “species 1, 2, 3” as “species 88, 12, 40” and lose nothing, because nothing but sameness was ever encoded.
- **Ordinal.** Numbers encode *order* but not spacing: pain scores, Likert items, tumor grades, disease stages. The admissible transformations are *any* order-preserving (monotone increasing) function. The rank is real; the size of the gap between ranks is not. Grade III sits above grade II, but “how much” above is not a fact the scale carries.
- **Interval.** *Differences* are meaningful, but there is no true zero: temperature in degrees Celsius, calendar date. The admissible transformations shrink to $x \mapsto ax + b$ with $a > 0$ (positive linear, i.e. rescale and shift). You may add and subtract meaningfully; you may *not* say 40° is “twice” 20°, because the zero is a convention, not an absence.
- **Ratio.** A true zero anchors the scale, so *ratios* are meaningful: length, mass, absolute temperature, molar concentration, counts. The admissible transformations shrink further to $x \mapsto ax$ with $a > 0$ (rescale only; the zero is fixed). Every arithmetic operation is licensed.

Look at the shape of the list, because the shape is the content. As you move *up* from nominal to ratio, the scale preserves *more* structure and the group of admissible transformations gets *smaller*.

As you move *down*, you *forget* structure and the group of transformations that leave you unharmed gets *larger*. This is precisely the “structure preservation” ladder of Chapter 3, now applied to measurement, and it is the same trade you saw when forgetting distance carried you from geometry down to topology in Chapter 9: fewer preserved relations, a bigger group of transformations that count as “not changing anything,” and correspondingly fewer statements that stay true across that bigger group. A larger symmetry group always means fewer invariants. Nominal scales are the topology of measurement, ratio scales its rigid geometry.

It is worth naming the four groups explicitly, because seeing them side by side makes the nesting undeniable and ties the whole scheme back to Chapter 7. The ratio scale’s group is the positive real numbers under multiplication (the rescalings $x \mapsto ax$). The interval scale’s group is the larger affine group ($x \mapsto ax + b$), which contains the ratio group as the special case $b = 0$. The ordinal scale’s group is larger still: *all* monotone increasing functions, an enormous group that contains every affine map with $a > 0$ and uncountably many more besides. The nominal scale’s group is the full group of one-to-one relabelings, the permutations of the label set, which contains everything monotone and more, since it need not respect order at all. Each group sits inside the next as you descend the ladder: positive scalings inside affine maps inside monotone functions inside permutations. The invariants shrink in lockstep, exactly as a nested tower of symmetry groups always forces its invariants to shrink. This is the Chapter 7 idea of a group as “the transformations that leave something unchanged,” pointed now at your data: each scale type simply *is* the set of number-shuffles its empirical structure cannot feel.

Worked example: sorting your own instruments. Take the quantities you handle and place each on the ladder by asking what empirical operation actually stands behind it.

- **Molar concentration:** ratio. There is a genuine zero (no molecules present) and you can meaningfully concatenate (mix two volumes). Every operation is licensed, including fold-changes.
- **Temperature in degrees Celsius:** interval. Differences are meaningful, but 0 °C is the freezing point of water, a convention, not the absence of thermal energy; so “twice as hot” is not meaningful. Switch to absolute temperature and the true zero returns, promoting it to ratio.
- **pH:** interval, and instructively so. It is defined as the negative base-10 logarithm of hydrogen-ion concentration, so equal *differences* in pH correspond to equal *fold-changes* in concentration, and pH 0 is 1 molar, not an absence of acidity. Differences are meaningful; ratios of pH values (“pH 6 is twice pH 3”) are not.
- **qPCR Ct value:** interval. Each cycle roughly doubles the template, so Ct behaves like a negative logarithm of the starting amount. A *difference* of two Ct values is meaningful (it reports a fold-change in template); a *ratio* of Ct values, or calling Ct 30 “twice” Ct 15, is not. Near the detection limit it degrades toward ordinal, because there the mapping to real template amount stops being trustworthy.
- **A 5-point Likert symptom score:** ordinal. Order is real; equal spacing between “mild,” “moderate,” and “severe” is an assumption the response format does not guarantee. The mean is, strictly, not licensed.
- **Tumor grade (I to IV):** ordinal. The grades rank severity; the gap from I to II need not equal the gap from III to IV. Order only.
- **Species identity:** nominal. Equality only. Order, mean, and every arithmetic operation are meaningless on it.

The two genuinely ambiguous cases (Ct near its limit, Likert treated as interval) are not defects in the taxonomy; they are the taxonomy telling you exactly where to look for trouble, and what extra information about the empirical operations would settle the classification. Hold that thought; it is the whole point of the corrective section below.

Meaningfulness: the invariance test

Now the connection to Chapter 5 turns into a rule you can apply every single day. Define it sharply: a statistical statement is **meaningful** exactly when its truth is *invariant* under the admissible transformations of the scales involved. If a re-expression of the numbers that the scale itself declares to be just as valid can *change* the truth of your statement, then your statement was never about the world; it was about your arbitrary choice of numbers. This is Noether’s logic transplanted into data analysis: what is real is what survives the transformations that merely change the description. In Chapter 5 the transformations were symmetries of a physical system and the invariants were conserved quantities; here the transformations are the admissible re-scalings of a measurement and the invariants are the statistics you are allowed to trust. Same idea, new domain.

Run Chapter 1’s averaging question through the test and watch it finally get a precise answer.

- The **mean** requires addition, so it is meaningful only for interval and ratio scales. Take the mean of ordinal pain scores coded 1 to 5. A monotone re-labeling of those categories, say 1, 2, 3, 4, 5 becomes 1, 2, 3, 4, 100, is admissible for an ordinal scale (it preserves the order, which is all the scale claims to encode) and yet it can *change which group has the higher average*. The two codings are equally valid representations of the identical empirical facts, and they disagree about your conclusion. That is the exact structural sense in which “averaging Likert scores fails”: the answer is an artifact of a choice the world never made.
- The **median** requires only order, so it *is* meaningful for ordinal data. A monotone transformation moves the numerical value of the median around, but it cannot change *which* group’s median is higher, because it preserves order and “higher” is an order-fact. The conclusion is invariant; the statistic is safe.
- **Ratios** (“twice as expressed,” a two-fold change) require a true zero, so they are meaningful only for ratio scales. This is why fold-changes are computed on raw concentrations and counts, which have a genuine zero, and never on interval quantities like a Celsius temperature or a pH.

Aside

The rule fits in a sentence you can carry into any analysis: *before trusting a statistic, ask what admissible transformations its scale allows, then ask whether the statistic’s verdict would survive them*. If it would, the statistic reports the world. If it would not, it reports your arbitrary choice of numbers, and no amount of downstream sophistication can launder it back into a fact.

You now have the vocabulary the Chapter 1 exercise gestured toward and could not supply. “When does averaging fail?” It fails, on measurement-theoretic grounds and before any question of sample size or tails, whenever the data are not *at least* interval, because then the mean is not invariant under the scale’s admissible transformations. Two entirely different failures wear the same clothes: the statistical one from Chapter 11 (the average does not concentrate) and the

measurement one from this chapter (the average is not invariant). A good analyst checks for both, and names which is which.

Worked example: the mean that reverses under a legal relabeling. A study reports the “mean tumor grade” of two treatment groups, codes the grades I to IV as 1 to 4, finds group A’s mean is 2.8 against group B’s 2.5, and concludes group A is worse. Grade is ordinal, so any monotone recoding of the four categories is admissible. Recode the grades instead as 1, 2, 3, 10, a perfectly legal order-preserving relabeling that merely says grade IV is much worse than III (which no one would dispute). If group B holds more grade-IV patients while group A holds more grade-III, the averages can now flip, with B coming out worse. Nothing about the patients changed; only an arbitrary numbering did, and the conclusion inverted with it. The conclusion was therefore never a fact about the patients. What the authors *should* have run is a rank-based comparison (a test of whether one group’s grades tend to be higher, or a comparison of medians or of the full grade distribution), because those verdicts are invariant under every admissible monotone relabeling and so report a property of the world rather than of the coding. That invariance is precisely why they are safe, and it is the invariance criterion, not statistical taste, that certifies them.

Worked example: why the log-fold-change is trustworthy and the mean pH is delicate. Raw concentration is a ratio scale: its admissible transformations are the pure rescalings $x \mapsto ax$, which form the group of positive reals under multiplication. Take the logarithm and something clean happens to that group. A rescaling $x \mapsto ax$ becomes, in log-space, an *added constant*: $\log(ax) = \log(x) + \log(a)$. So the log turns the multiplicative group of admissible transformations into an *additive* one, which is exactly the signature of an interval scale. Log-concentration is an interval scale, and on an interval scale *differences* are meaningful. A log-fold-change is a difference of two log-concentrations, hence the logarithm of a ratio of concentrations, and it is meaningful for both reasons at once: it is a licensed difference on the log (interval) scale and a licensed ratio on the raw (ratio) scale. “A log-fold-change of 2” is a solid, invariant statement. Now contrast the “mean of two pH values.” pH is already a logarithm of concentration, so it is interval, and the mean of an interval scale *is* technically licensed (it commutes with the affine admissible transformations, so which group has the higher mean pH is invariant). The delicacy is not that the operation is forbidden; it is that averaging pH is averaging *logarithms*, which is the geometric mean of the underlying hydrogen-ion concentrations, not the arithmetic mean. So “mean pH” is a meaningful statistic about the pH scale that answers a *different question* from the one you probably think you are asking about average acidity. The scale licenses the operation; it does not license your intuition about what the result means.

The corrective: do not weaponize the taxonomy

Everything above could be misread as a rulebook, a set of prohibitions to enforce on colleagues, and the reading you are doing alongside this chapter exists specifically to stop that. Velleman and Wilkinson’s point is that scale type is *not* an intrinsic property stamped onto data at birth. It depends on the *question you are asking* and on the *empirical operations that actually stand behind the numbers*, and both of those can shift.

Two directions of drift make the point. Ordinal-looking data are frequently, and often justifiably, analyzed as interval. Many well-constructed rating scales approximate equal spacing closely enough that treating them as interval loses little and gains the whole toolbox of means and linear models; the approximation is an empirical claim you can check, not a sin. Conversely, ratio-looking counts can behave ordinally near a detection limit, where the mapping from number to reality goes soft and only the ranking survives; there the interval and ratio structure you would naively assume is not actually there.

Worked example: the same move, justified once and fatal once. Consider two uses of “treat ordinal as interval.” First, a validated quality-of-life questionnaire whose many items, summed, have been calibrated against external anchors so that a one-unit change means roughly the same thing across the range. Averaging those summed scores across a large group is defensible: the spacing is approximately real (an empirical fact the calibration established), the aggregate concentrates by Chapter 11, and a monotone relabeling drastic enough to overturn a conclusion would have to contradict the calibration data. Here interval-treatment gains you power at negligible cost. Second, a single four-point clinician-assigned severity grade, averaged across a handful of patients to declare one arm worse. The spacing here was never established (is the step from grade III to IV really equal to the step from I to II?), the sample is small so nothing concentrates, and, as the tumor-grade example showed, a legal monotone relabeling can flip the verdict. Same operation, opposite verdicts. What distinguishes them is not the label “ordinal”; it is whether the equal-spacing assumption is *backed by empirical evidence* and whether the conclusion is *robust to admissible relabeling*. Velleman and Wilkinson’s corrective and the invariance test are not in tension: the test tells you exactly what evidence the interval-treatment needs, and the corrective reminds you that such evidence sometimes exists. The taxonomy, used well, is not a gate that forbids operations. It is a *lens* that makes one question sharp: *which operations are invariant here, given what these numbers really came from?* Its value is that it lets you notice when an analysis is quietly riding on structure the measurement does not carry, so you can ask whether that structure is really present or merely assumed. A tool for noticing, not a rule for forbidding. The colleague who says “you must *never* average ordinal data” has turned the lens into a cudgel and will sometimes be wrong; the condition under which they are right is precise, and it is the one the invariance test names: when the spacing genuinely is not there, and a legal monotone relabeling could overturn your conclusion, the average is an artifact and must go.

Where this sits on your map

Step back and see the whole pipeline. Your data are assays, scores, and proxies: Ct values, pH readings, Likert ratings, log-fold-changes, ordinal severity grades. Every one of them is a homomorphism from some empirical relational structure into the numbers, and every one carries a specific, checkable amount of structure that licenses some operations and forbids others. Measurement is the daily-relevant *first* step of the pipeline that runs numbers $\rightarrow$ structure $\rightarrow$ prediction. It is the step that manufactures the numbers on which every later step operates, which is exactly why getting it wrong is so costly: the error is not noise that averaging can beat down. It is a *category mistake* about what the numbers mean, and every downstream model inherits it intact, no matter how sophisticated the model or how large the sample. You cannot out-compute a measurement that was never a faithful homomorphism to begin with.

Where you now stand

You came into this chapter with numbers you treated as given, and a vague sense that “some averages are meaningless” without a way to say which or why. You leave with a precise account. A scale is a homomorphism from an empirical relational structure to a numerical one; its rigidity is captured by its group of admissible transformations; the four scale types are four rungs of that rigidity, from nominal (largest group, least structure) to ratio (smallest group, most structure); and a statistic is meaningful exactly when it is invariant under that group. This is the same invariance principle from Chapter 5, now doing daily work, and it completes the answer to the Chapter 1 averaging question that Chapter 11 began: averaging can fail statistically (no concentration) or

measurement-theoretically (no invariance), and those are different diagnoses with different fixes.

The chapter hands one thing forward, and it is the keystone of the next. You have now watched the *first* homomorphism in the pipeline, the one that turns a piece of the world into numbers. The next chapter claims that a *model* is the very same kind of object, a structure-preserving map, running one step later: from the numbers, or from the system they describe, into a mathematical structure with its own internal laws. If measurement is a homomorphism and a model is a homomorphism, then the entire act of applying mathematics to reality is a *chain* of structure-preserving maps, and the reasons a model works, and the three specific ways it fails, are about which structure each map in the chain preserves and which it drops. That is Chapter 14, where “what a model is” finally gets stated in full, using the map you built here as its first link.